# PROJECT AND TEAM INFORMATION

## Project Title

*Real-Time Driver Drowsiness and Fatigue Tracking System Using Machine Learning*

## Student / Team Information

| | |
|---|---|
| *Team name* | *IML-V-2026-T045* |
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# PROPOSAL DESCRIPTION

## Motivation

*Drowsy driving and operator fatigue are leading contributors to severe traffic accidents, highway fatalities, and workplace equipment injuries globally.*

*An proactive safety monitor that continuously tracks driver attentiveness can prevent accidents before they happen. By utilizing non-invasive computer vision to analyze facial cues in real time, we can assess fatigue indicators such as micro-sleeps and continuous yawning. Building an accessible, low-latency safety system provides drivers and fleet managers with immediate warnings, reducing road hazards and saving lives.*

## State of the Art / Current solution

*Existing driver monitoring solutions primarily fall into two categories: high-end hardware sensors and simple threshold-based software applications.Software-based solutions frequently use simple, hardcoded threshold rules (such as triggering an alert if eye openness drops below a fixed percentage).*

*These rule-based methods generate high rates of false alarms because they fail to account for facial variation, glasses, or natural lighting changes. Our approach replaces hardcoded rules with a supervised Machine Learning pipeline trained on multi-feature facial metrics to deliver robust, intelligent fatigue predictions.*

## Project Goals and Milestones

*The main goal of this project is to build an end-to-end Machine Learning driver safety monitor that analyzes facial features in real time, classifies fatigue states, and triggers immediate alarms.*

***Milestone 1: Landmark Processing & Dataset Engineering***

*Extracts 468 3D facial landmarks from benchmark driver videos (e.g., UTA-RLDD) using MediaPipe. Calculates spatial Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), rolling averages, and blink frequencies into a clean structured .csv dataset.*

***Milestone 2: ML Model Training & Optimization***

*Trains, tunes, and evaluates Random Forest, SVM, and Logistic Regression algorithms on the extracted features. Optimizes for >90% recall on the Critical Drowsiness class to eliminate false negatives, serializing the best model as a .pkl file.*

***Milestone 3: Active Safety Alarms & PDF Reports***

*Triggers immediate multi-stage safety alerts (auditory alarms + flashing screen overlays) when fatigue is detected across rolling frames. Logs timestamped event data to SQLite and generates downloadable PDF shift reports.*

## Project Approach

*Our system uses a lightweight, modular computer vision and machine learning pipeline executed in Python:*

***Video Stream & Facial Landmarks:*** *OpenCV captures live webcam frames. MediaPipe Face Landmarker tracks 468 3D facial coordinates on the driver's face in real time.*

***Feature Extraction:*** *Mathematical functions convert landmark coordinates into numerical vectors: Eye Aspect Ratio (EAR) for eye closure, Mouth Aspect Ratio (MAR) for yawning, and rolling averages over time windows.*

***Machine Learning Classification:*** *Features are passed to a pre-trained* ***Random Forest / SVM classifier*** *(serialized as a .pkl file using Joblib). The model classifies the driver's current state as* ***Alert (0)****,* ***Mild Fatigue (1)****, or* ***Critical Drowsiness (2)****.*

***Interface & Alerts:*** *The predicted state updates a clean dashboard interface built with HTML/CSS/JavaScript (or OpenCV canvas). If Critical Drowsiness is detected across continuous frames, audio-visual alarms trigger instantly.*

## System Architecture

*1. INPUT MODULE*
*Webcam Video Stream (OpenCV)*
| *Raw RGB Frames*
∨
*2. COMPUTER VISION PIPELINE*
*MediaPipe 468 Face Landmarks*
| *Scaled Feature Vector[ EAR, MAR and Blink Variance]*
∨
*3. MACHINE LEARNING INFERENCE*
*Feature Standardization (StandardScaler)*
*Supervised Classifier (Random Forest / SVM)*
*Class Output:*
*0 - Alert | 1 - Mild | 2 - Drowsy*
| *State Code & Metrics Stream*
∨
*4. APPLICATION INTERFACE*
*HTML5 / CSS3 / JavaScript Dashboard*
*Real-Time Charts (Chart.js)*
*Active Alarms*
*Timestamp Event Logging (SQLite)*

## Project Outcome / Deliverables

*An end-to-end Python safety application that monitors live driver video at 30+ FPS, evaluates fatigue using serialized ML models (.pkl), triggers real-time audio-visual alarms during critical drowsiness, and generates performance reports detailing model metrics like recall and confusion matrices.*

## Assumptions

*The camera operates with an unobstructed, direct line of sight to the driver's face under standard ambient vehicle lighting.*

*Natural facial variations (such as smiling or talking) are treated as transient noise that the multi-feature ML model filters out.*

## References

*Soukupová, T., & Cech, J. (2016). Real-Time Eye Blink Detection Using Facial Landmarks. Center for Machine Perception, Czech Technical University in Prague.*

*Ghoddoosian, R., Galib, M., & Selim, M. (2019). A Realistic Dataset and Baseline Method for Drowsiness Detection (UTA-RLDD). IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops.*

*Lugaresi, C., et al. (2019). MediaPipe: A Framework for Building Perception Pipelines. Google Research.*